

# Chapter 18: Smaller Ecdysozoans

## Zoology

Instructor

November 28, 2025

# Outline

**Introduction to Ecdysozoa**

**Phylum Nematoda**

**Phylum Nematomorpha**

**Phylum Kinorhyncha**

**Phylum Priapulida**

**Phylum Loricifera**

**Clade Panarthropoda**

# Outline

## Introduction to Ecdysozoa

### Phylum Nematoda

### Phylum Nematomorpha

### Phylum Kinorhyncha

### Phylum Priapulida

### Phylum Loricifera

### Clade Panarthropoda

# Ecdysozoa Overview

- **Protostome** animals are divided into **Lophotrochozoa** and **Ecdysozoa**.
- **Ecdysozoa** (蜕皮动物) comprises taxa that molt their cuticle as they grow.
- **Cuticle** (角质层): A nonliving external layer secreted by the epidermis.
- **Ecdysis** (蜕皮): The process of shedding the outer layer of the cuticle.
- Molting is regulated by the hormone **ecdysone** (蜕皮激素).
- Ecdysozoan phyla include:
  - **Nematoda** (线虫动物门)
    - **Nematomorpha** (线形动物门)
    - **Kinorhyncha** (动吻动物门)
    - **Priapulida** (鳃曳动物门)
    - **Loricifera** (铠甲动物门)
    - **Onychophora** (有爪动物门)
    - **Tardigrada** (缓步动物门)

# Ecdysozoa Overview

- **Arthropoda** (节肢动物门)

# Ecdysozoan Phylogeny

- **Nematoidea** (线虫动物总门): Nematoda + Nematomorpha.
- **Panarthropoda** (泛节肢动物): Onychophora + Tardigrada + Arthropoda.
- **Scalidophora** (有棘动物): Kinorhyncha + Priapulida + Loricifera.

# Phylogeny Cladogram

- Cladogram depicting relationships among ecdysozoan phyla.
- Shows the division into **Nematoidea** (Nematoda + Nematomorpha) and **Panarthropoda** (Onychophora + Tardigrada + Arthropoda).
- Key characters include molting cuticle, reduction of coelom, and specific morphological traits.



# Outline

Introduction to Ecdysozoa

**Phylum Nematoda**

Phylum Nematomorpha

Phylum Kinorhyncha

Phylum Priapulida

Phylum Loricifera

Clade Panarthropoda



# Phylum Nematoda: Roundworms

- Approximately 25,000 species named; estimates up to 500,000.
- Found in sea, freshwater, soil, and as parasites of plants and animals.
- **Free-living nematodes** feed on bacteria, yeasts, fungal hyphae, algae, or are predatory.
- **Parasitic nematodes** affect virtually every type of animal and many plants.
- **Caenorhabditis elegans**: An important model organism for developmental biology and genetics.

# Form and Function

- Cylindrical shape; flexible, nonliving **cuticle**.
- Lack motile cilia or flagella (except in one species).
- **Eutely** (细胞定数): Constant number of cells.
- **Muscles**: Longitudinal only; no circular muscles.
- **Excretory system**: Gland cells or canal system; no protonephridia.
- **Pharynx**: Muscular with a triradiate lumen.
- **Cuticle**: Secreted by the underlying epidermis (**hypodermis**).
  - Composed primarily of **collagen**.
  - Contains high hydrostatic pressure (**turgor**).

# Nematode Structure

- Structure of a nematode as illustrated by **Ascaris** female.
- **Ascaris** has two ovaries and uteri, which open to the outside by a common genital pore.
- Cross section shows longitudinal muscles and **pseudocoel** (假体腔).
- Single muscle cell structure: spindle abuts hypodermis, muscle arm extends to nerve.



# Nematode Sensory Organs

- Diagram of an **amphid** (头感器) in ***Caenorhabditis elegans***.
- Amphids are complex sensory organs on the head (chemoreceptive).
- **Phasmids** (尾感器) are sensory organs near the posterior end in parasitic nematodes.



# Common Parasitic Nematodes

- Table listing common parasitic nematodes of humans in North America.
- Includes Hookworm, Pinworm, Intestinal roundworm (**Ascaris**), Trichina worm, and Whipworm.
- Lists mode of infection and prevalence.





# Nematode Reproduction

- Cross section of a male nematode showing reproductive structures.
- Posterior end of a male nematode showing **copulatory spicules** (交合刺).
- Males are smaller than females.
- Fertilization is internal.
- Sperm lack flagella and acrosome; amoeboid movement.



# Ascaris lumbricoides

- **Ascaris lumbricoides** (似引蛔线虫), the large roundworm of humans.
- Male is smaller with a curved tail; female is larger.
- Intestine of a pig blocked by **Ascaris suum**.
- Heavy infections can cause intestinal blockage.



# Ascaris Life Cycle

- Eggs passed in feces; develop in soil (2 weeks).
- Infection via ingestion of embryonated eggs (contaminated food/soil).
- Juveniles hatch in intestine, burrow into veins/lymph vessels.
- Carried to heart and lungs; break into alveoli.
- Migrate up trachea, swallowed, return to intestine.
- Mature in intestine (2 months after ingestion).



# Hookworms

- Mouth of hookworm displaying cutting plates.
- Section through anterior end of hookworm attached to dog intestine.
- Cutting plates pinch off mucosa; muscular pharynx sucks blood.
- Esophageal glands secrete anticoagulant.



# Hookworm Life Cycle

- Life cycle of hookworms (***Necator americanus***, ***Ancylostoma duodenale***).
- Eggs passed in feces; juveniles hatch in soil.
- Infective juveniles burrow through human skin (contact with soil).
- Migrate to lungs via blood, then to intestine.



# Trichina Worm

- Muscle infected with trichina worm **Trichinella spiralis** (旋毛虫).
- Juveniles lie within muscle cells transformed into **nurse cells** (保姆细胞).
- Infection via eating raw/undercooked meat (pork) containing encysted juveniles.
- Can infect humans, hogs, rats, cats, dogs.





# Pinworms

- Pinworms, **Enterobius vermicularis** (蛲虫).
- Female worm from human large intestine.
- Group of pinworm eggs.
- Females migrate to anal region at night to lay eggs, causing itching.
- Infection via ingestion of eggs.



# Filarial Worms: **Wuchereria bancrofti**

- Life cycle of **Wuchereria bancrofti** (班氏吴策线虫).
- Transmitted by mosquitos.
- Juveniles migrate to lymph nodes and develop to maturity.
- Adult worms produce **microfilariae** (微丝蚴) which enter blood circulation.



# Elephantiasis

- **Elephantiasis** (象皮肿) of leg caused by adult filarial worms.
- Worms block the flow of lymph, causing excessive growth of connective tissue and swelling.





# Dog Heartworm

- ***Dirofilaria immitis*** (犬恶丝虫) in right ventricle of a dog heart.
- Extends into pulmonary arteries.
- Transmitted by mosquitos.
- Serious disease for dogs.





# Outline

Introduction to Ecdysozoa

Phylum Nematoda

**Phylum Nematomorpha**

Phylum Kinorhyncha

Phylum Priapulida

Phylum Loricifera

Clade Panarthropoda

# Phylum Nematomorpha: Horsehair Worms

- Structure of **Paragordius**, a nematomorph.
- “Horsehair worms” (铁线虫); superficially resemble hairs.
- Adults free-living in moist habitats; juveniles parasitic in arthropods.
- Vestigial digestive system; absorb nutrients through body wall.



# Outline

Introduction to Ecdysozoa

Phylum Nematoda

Phylum Nematomorpha

**Phylum Kinorhyncha**

Phylum Priapulida

Phylum Loricifera

Clade Panarthropoda

# Phylum Kinorhyncha

- **Echinoderes**, a kinorhynch (动吻动物).
- Minute marine worms, “spiny necked”.
- Body divided into head, neck, and trunk (11 segments).
- Retractable head (**introvert**) with circlets of spines (**scalids**).
- Burrow in mud/silt.



# Outline

Introduction to Ecdysozoa

Phylum Nematoda

Phylum Nematomorpha

Phylum Kinorhyncha

**Phylum Priapulida**

Phylum Loricifera

Clade Panarthropoda



# Phylum Priapulida

- Major internal structures of **Priapulus** (鳃曳动物).
- **Priapulus caudatus** from Lurefjord, Norway.
- Cylindrical body: **introvert**, **trunk**, and **caudal appendages**.
- Burrowing predators or detritus feeders in cold marine waters.



# Outline

Introduction to Ecdysozoa

Phylum Nematoda

Phylum Nematomorpha

Phylum Kinorhyncha

Phylum Priapulida

**Phylum Loricifera**

Clade Panarthropoda

# Phylum Loricifera

- Adult loriciferan, **Nanaloricus mysticus** (铠甲动物).
- Tiny animals living in marine gravel.
- Protective external case (**lorica**).
- Introvert with scalids retracts into lorica.



# Outline

Introduction to Ecdysozoa

Phylum Nematoda

Phylum Nematomorpha

Phylum Kinorhyncha

Phylum Priapulida

Phylum Loricifera

**Clade Panarthropoda**

# Phylum Onychophora: Velvet Worms

- **Peripatus**, a caterpillar-like onychophoran (有爪动物).
- “Velvet worms” or “walking worms”.
- Tropical/subtropical moist habitats.
- Characteristics in common with both annelids and arthropods.





# Onychophora Anatomy

- Internal anatomy of an onychophoran.
- **Slime glands** (黏液腺) expel sticky fluid for defense.
- **Tracheal system** (气管系统) for respiration; spiracles cannot close.
- Open circulatory system; dorsal tubular heart.
- Ladder-like nervous system.





# Phylum Tardigrada: Water Bears

- Scanning electron micrograph of an aquatic tardigrade, **Pseudobiotus** (缓步动物).
- “Water bears”; minute ( $< 1$  mm).
- Mostly terrestrial (water film of mosses/lichens).
- Four pairs of short, stubby, **unjointed legs** with claws.



# Tardigrada Anatomy

- Internal anatomy of a tardigrade.
- Mouth with **buccal tube** and **stylets** (piercing/sucking).
- **Malpighian tubules** (马氏管) for excretion.
- No circulatory or respiratory systems.
- **Cryptobiosis** (隐生): State of suspended animation to withstand harsh conditions.



# Tardigrada Reproduction

- Dioecious (some parthenogenetic).
- Eggs are often ornate.
- Molting occurs periodically.
- Females of some species deposit eggs in the molted cuticle.

# Tardigrada Eggs

- Scanning electron micrograph of a highly ornate egg of a tardigrade, **Macrobiotus hufelandii**.
- Eggs are often ornate.





# Tardigrada Molting

- Molted cuticle of a tardigrade, containing a number of fertilized eggs.
- Females of some species deposit eggs in the molted cuticle.

